

B.Tech. Degree III Semester Examination November 2013

ME 1306 MACHINE DRAWING (2012 Scheme)

(Missing dimensions if any may be assumed)

Time: 4 Hours

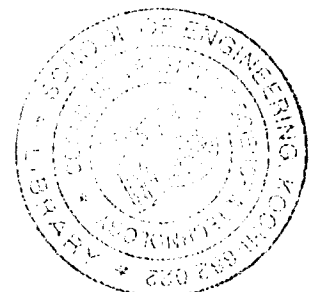
Maximum Marks : 100

PART A (Answer *ALL* questions)

- I. (a) Draw profile of BSW thread showing at least three threads. $P=36\text{mm}$ and indicate all the standard proportions in terms of P . (10)
- (b) Make a neat sketch of both the external and internal square thread taking $P=24\text{mm}$. Show at least three threads. Indicate all the proportions in the drawing. (10)
- (c) Draw neat dimensional sketch of Lewis foundation bolt having a diameter of 20mm at the threaded end. Indicate all the proportions in the drawing. (10)
- OR**
- II. Details of a knuckle joint is shown in Fig.1. Assemble them and draw: (30)
- (a) elevation (b) plan top half in section (c) side view from left
- III. Draw top half sectional elevation of an integral flanged joint for the following specifications: size of the pipe to be joined = $\phi 80$, Outside diameter of the pipe = 100, outside diameter of the flange = 176. Pitch circle diameter of bolts = 140. Size of the bolt = M12. Number of bolts = 6. Thickness of the flange = 20. Thickness of the gasket = 3. All dimensions are in millimeter. Also draw an end view. (30)
- OR**
- IV. Isometric view of a flanged coupling (protected type) is shown in fig.2. Draw the top half sectional elevation and an end view. (30)
- V. Fig.3 shows details of a steam stop valve. Assemble and draw: (40)
- (a) Right half sectional elevation.
(b) Top half sectional plan, taking section along the centre line of the outlet of the valve.
- OR**
- VI. Fig.4 shows the details of a Lathe tailstock. Assemble the parts and draw : (40)
- (a) sectional elevation
(b) plan

(Figures overleaf)

(P.T.O.)



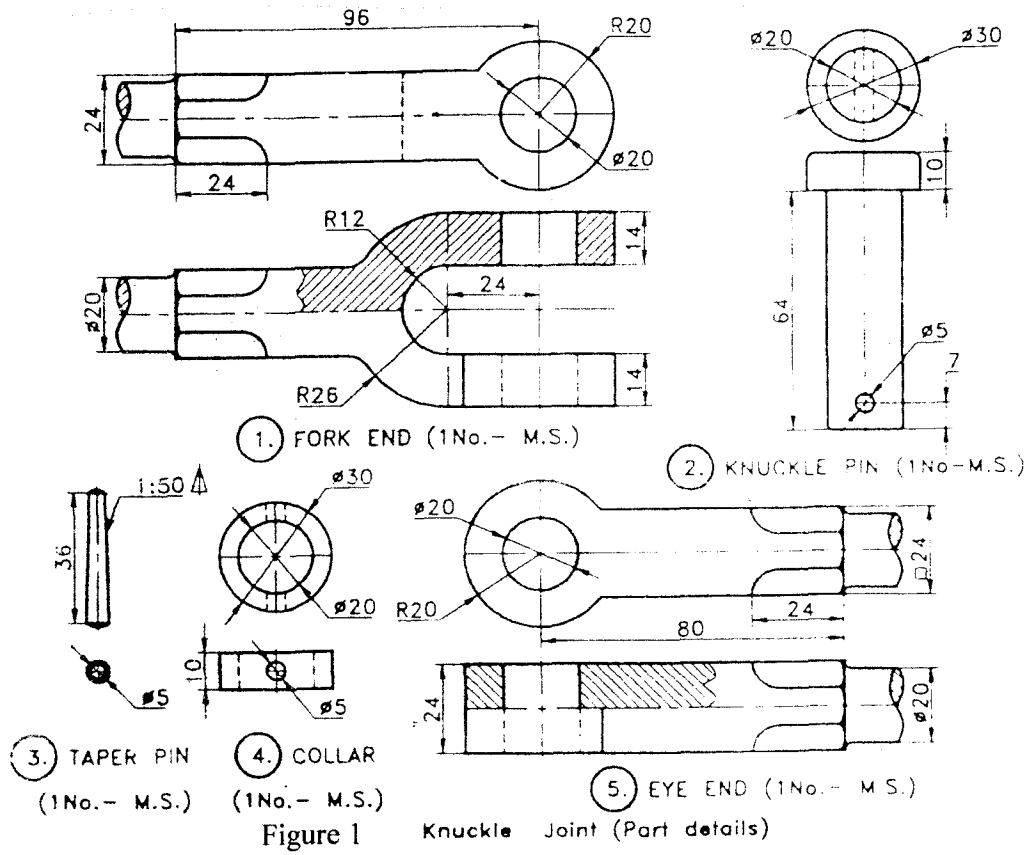


Figure 1

Knuckle Joint (Part details)

Flanged Coupling (Protected type)

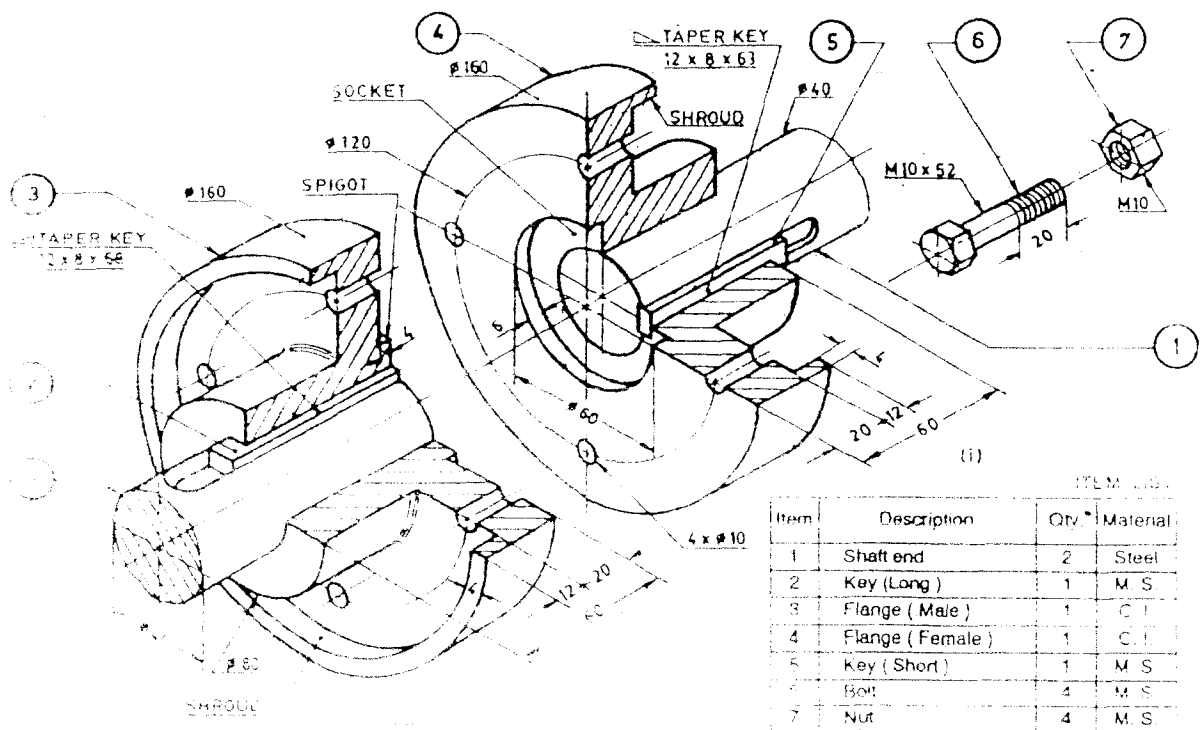


Figure 2

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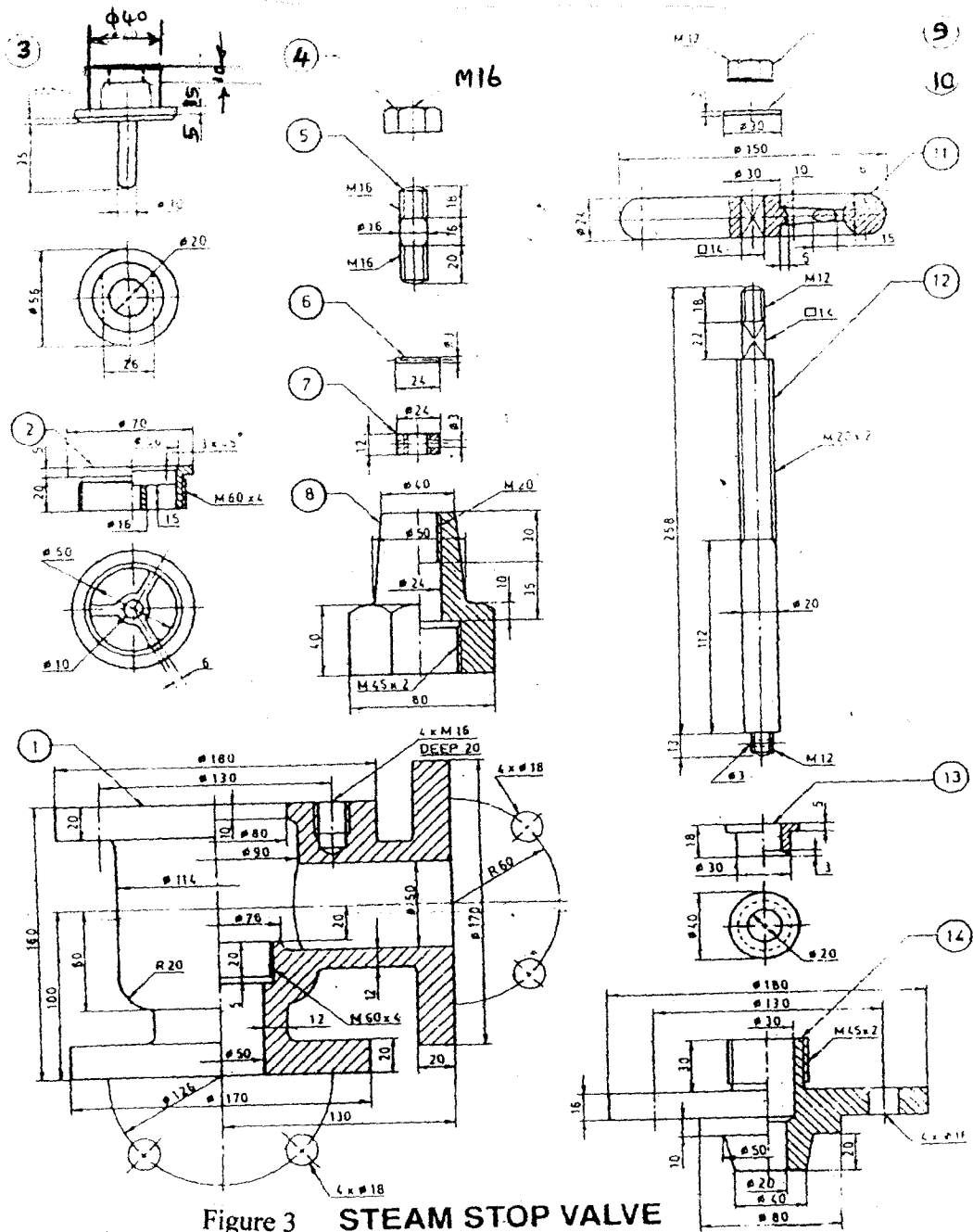
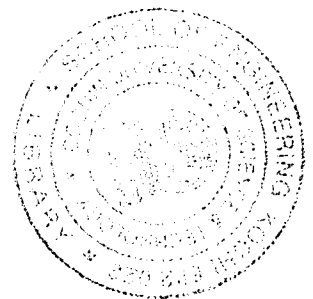


Figure 3 STEAM STOP VALVE

ITEM LIST

Item	Description	Qty	Material
1	Body	1	C.I.
2	Valve Seat	1	G.M.
3	Valve	1	G.M.
4	Nut for stud	4	M.S.
5	Stud	4	M.S.
6	Pin for collar	1	G.M.
7	Collar nut	1	G.M.
8	Gland nut	1	M.S.
9	Nut for spindle	1	M.S.
10	Washer	1	M.S.
11	Hand wheel	1	C.I.
12	Spindle	1	M.S.
13	Gland bush	1	G.M.
14	Cover	1	C.I.



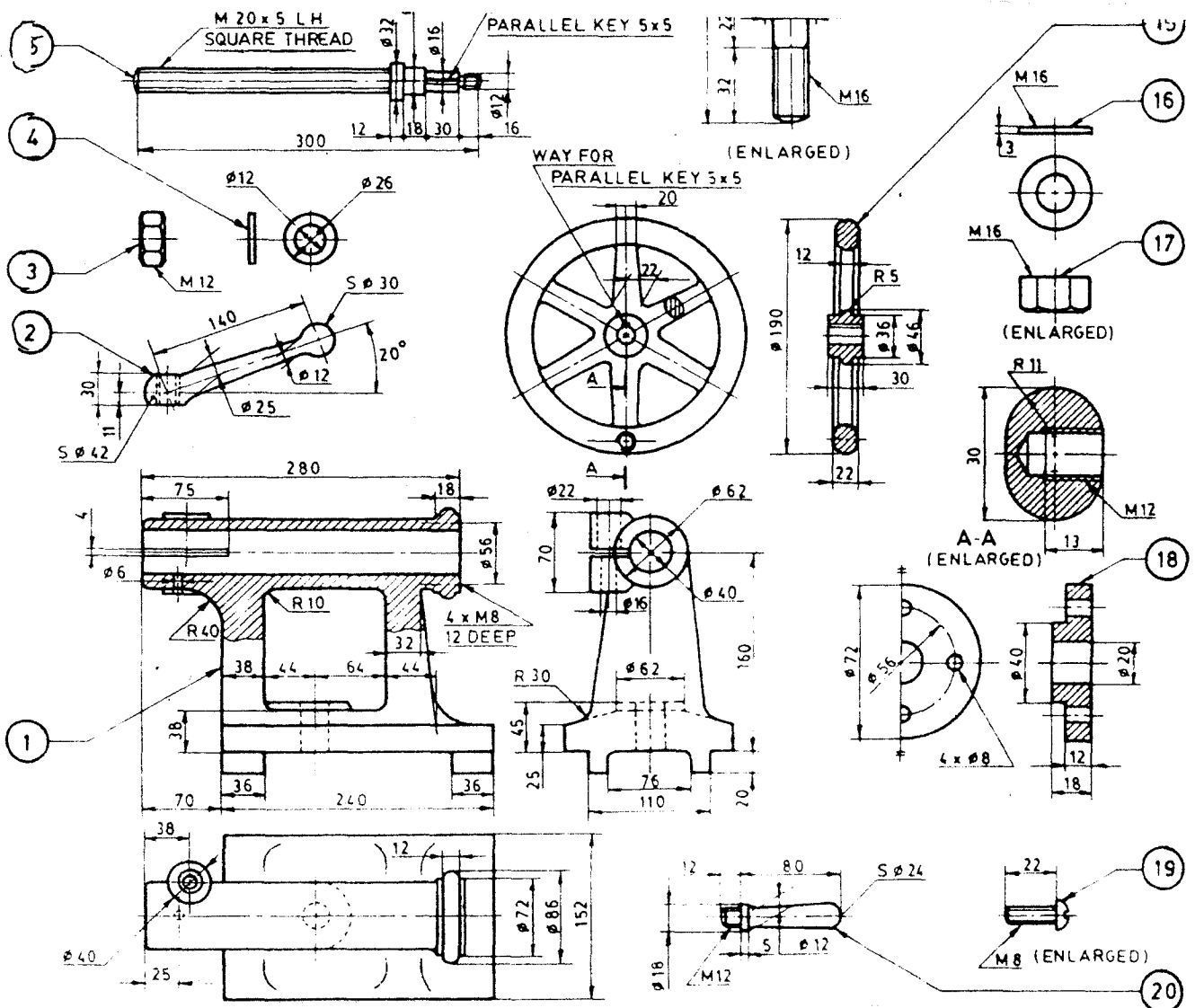


Figure 4 Tail Stock Parts

ITEM LIST

Item	Description	Qty.	Material
1	Body	1	C. I.
2	Handle (Lock)	1	C. I.
3	Nut (M 12)	1	M. S.
4	Washer (M 12)	1	M. S.
5	Screw spindle	1	M. S.
6	Clamping plate	1	Steel
7	Dead centre	1	Steel
8	Nut (M 22)	1	M. S.
9	Sq. bolt (M 22)	1	M. S.
10	Washer (M 22)	1	M. S.

Item	Description	Qty.	Material
11	Barrel	1	C. I.
12	Key (Barrel)	1	M. S.
13	Stud	1	M. S.
14	Key (Wheel)	1	M. S.
15	Hand Wheel	1	C. I.
16	Washer (M 16)	1	M. S.
17	Nut (M 16)	1	M. S.
18	Flange	1	C. I.
19	Screw (M 8)	4	M. S.
20	Handle (Wheel)	1	M. S.